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Efficient dual-branch high-resolution transformer design for crack segmentation

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HIGHLIGHTS

- An efficiently designed dual-branch high-resolution transformer model for crack segmentation.
- A multi-scale spatial-reduction attention mechanism is incorporated in the high-resolution branch for detail extraction.
- An enhanced ReLU-based linear attention mechanism is implemented in the low-resolution branch for learning global features.
- A bidirectional cross-resolution feature enhancer is introduced to effectively integrate multi-scale features across different branches.

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ABSTRACT

Recent studies show that Transformer-based crack segmentation models deliver impressive results. However, due to the quadratic complexity of the self-attention mechanism, the computational cost becomes prohibitive when handling high-resolution dense prediction tasks. This paper presents an efficient Transformer-based network for crack segmentation, utilizing several specially optimized self-attention modules. The proposed model incorporates a dual-branch structure, consisting of a high-resolution branch that maintains high resolution throughout and a low-resolution branch that progressively reduces resolution. A multi-scale spatial-reduction attention is introduced to the high-resolution branch to capture fine details, while an enhanced ReLU-based linear attention in the low-resolution branch learns global features. A bidirectional cross-resolution feature enhancement module connects the two branches, facilitating comprehensive information fusion. On the combined dataset CrackSeg9k and the scenario-specific datasets Asphalt3k and Concrete3k, the proposed method obtains mIoU of 81.85 %, 81.29 %, and 86.36 %, respectively, which outperforms the comparative state-of-the-art models and also achieves the best trade-off between efficiency and performance. Additionally, a robustness benchmark on common corruptions shows the proposed model degrades less than comparable lightweight baselines.

1. Introduction

In civil engineering, the appearance of cracks is often a precursor to structural damage, making early recognition and assessment of cracks crucial [1]. Traditional crack detection methods rely on manual visual inspection, which is not only time-consuming and labor-intensive but also prone to subjective judgment, leading to inconsistent detection results. With the rapid development of computer vision, automated crack detection technology has emerged to revolutionize structural health monitoring.

Automated crack detection technology typically leverages functional image acquisition devices and advanced image processing algorithms to achieve rapid crack recognition and precise measurement [2,3]. The current trend is to design automated detection systems integrated with advanced artificial intelligence (AI) algorithms [4,5]. These systems are capable of automatically identify crack characteristics, including length, width, and depth, and even assess the rate and trend of crack development. The application of AI technology can significantly enhance the efficiency and accuracy of crack detection, while reducing the interference of human factors.

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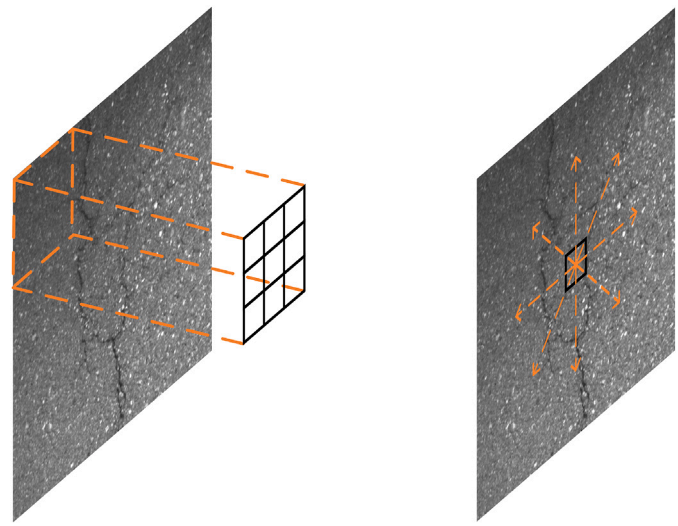
Currently, convolutional neural networks (CNNs) dominate intelligent crack detection methods, with their success primarily attributed to the inherent spatial inductive biases, namely locality and translation invariance [5]. These characteristics enable CNNs to efficiently learn and extract local features from images, making them particularly effective in crack detection tasks. By focusing on local regions of the image, CNNs can accurately identify and locate cracks, maintaining high performance regardless of variations in crack morphology and position. Although CNNs have demonstrated excellent performance in crack detection, they still face several challenges and limitations. Firstly, to improve computational efficiency and reduce the number of parameters, modern CNNs typically use small convolutional kernels, such as 3×3 or 5×5 [6]. While such design effectively captures fine local features, it limits the model's ability to capture global context. As a result, CNNs perform less effectively in modeling large-scale global information, making it difficult to fully leverage the structural information of the entire image for more accurate predictions. Secondly, the locality of CNNs makes them more sensitive to noise in images [7,8]. Since CNNs primarily rely on the relationships between pixels in local regions, the presence of noise or interference in an image can lead the model to incorrectly identify these pixels as meaningful features, thus compromising accuracy. This sensitivity to noise not only diminishes the model's robustness but also limits its generalization ability, meaning that the model may underperform when dealing with unseen or differently distributed data.

The emergence of Transformer has effectively addressed the limitations of CNNs, with its widely adopted self-attention mechanism proving highly effective in handling long-range dependencies and inherently possessing a global receptive field [9]. A comparison between CNN and self-attention is shown in Fig. 1. The introduction of Vision Transformer (ViT) has further demonstrated the significant potential of the self-attention mechanism in computer vision [10]. However, the computational complexity of the self-attention mechanism scales quadratically with the input sequence length, making the computational cost prohibitive when dealing with high-resolution features [11,12]. Despite the remarkable advantages of Transformer in capturing global features and managing long-range dependencies, its high computational and memory demands limit its practical application in high-resolution dense prediction.

Based on the above observations, there is clearly a significant gap between Transformer-based models and the accurate and efficient crack detection, making these models impractical for real-world applications. To bridge this gap, this paper proposes an Efficient Crack Transformer (ECFormer). ECFormer adopts a dual-branch structure, with the core innovation being the use of a customized efficient attention mechanism for each branch. This design endows the model with both a global receptive field and multi-scale learning capability, which are crucial for crack segmentation. Compared to pure-Transformer models, ECFormer significantly improves computational efficiency while maintaining performance. Specifically, a multi-scale spatial-reduction attention mechanism is designed to capture crack features at different scales in the high-resolution branch, while an enhanced ReLU-based linear attention mechanism with a global receptive field is applied in the low-resolution branch to extract contextual crack information. To facilitate comprehensive information exchange between the two branches, a bidirectional cross-resolution feature enhancer bridges the two branches, enabling thorough communication. The effectiveness of ECFormer is demonstrated through its performance on multiple crack segmentation benchmarks, and comparisons with state-of-the-art models show that it greatly reduces computational cost while preserving performance. The contributions of this paper are summarized as follows:

- A high-resolution dual-branch Transformer-based model called ECFormer is proposed for efficient crack segmentation tasks.
- In the ECFormer's high-resolution branch, a multi-scale spatial-reduction attention mechanism is designed to extract detailed features, while in the low-resolution branch, an enhanced ReLU-based linear attention mechanism is developed to capture global features. Additionally, a bidirectional cross-resolution feature enhancer is designed to facilitate information exchange between the two branches.
- ECFormer achieves the best performance compared to the state-of-the-art models on the combined dataset CrackSeg9k and the scenario-specific datasets Concrete3k and Asphalt3k, and achieves the best trade-off between performance and efficiency.
- A focused robustness evaluation benchmark on CrackSeg9k, covering Gaussian noise, motion blur, brightness shifts, and JPEG compression, with five severities and an aggregate robustness score.



(a) CNN (b) Self-attention

Fig. 1. Comparison of CNN and self-attention.

2. Related work

There are three lines of work related to this study. The first is general research on crack segmentation, which mainly includes rule-based and data-driven methods. The second focuses on crack segmentation based on Transformers, and the third involves improvement efforts aimed at addressing the limitations of Transformer.

2.1. Generic crack segmentation

The essence of image segmentation is to partition an image into multiple regions with consistency and coherence, thereby extracting the semantic information of the target objects or scenes and achieving a structured understanding of the image content. Crack segmentation is a specific application of image segmentation, aiming to extract crack regions from an image and distinguish them from the background. Over the past decade, crack segmentation has made substantial progress, evolving from early global threshold-based techniques to advanced deep learning algorithms. Based on the decision-making criteria, existing crack segmentation methods can be broadly categorized into two groups, rule-based methods and data-driven methods.

Rule-based methods rely on predefined rules, algorithms, or domain knowledge to achieve crack segmentation. These methods typically employ thresholding, edge detection, region growing, or clustering techniques, dividing regions based on conditions such as pixel intensity, color characteristics, or geometric features of the image.

Among rule-based approaches, thresholding and edge-based methods are the most commonly used due to their simplicity and intuitive nature. When the intensity distribution between crack and background pixels is significantly different, a global threshold can be applied to the entire image, known as the global thresholding method [13–17]. The main advantages of the global thresholding method are simplicity and computational efficiency. However, it fails when the assumption of two peaks in the histogram is violated. This is because, while the histogram provides quantitative information about the pixel intensity distribution, it neglects the spatial relationships and structural information between pixels, which are crucial for understanding and analyzing image content. To overcome this limitation, many studies have adopted methods that set thresholds based on local information [18,19]. These approaches not only consider the pixel's intensity but also its statistical properties within the local neighborhood, such as the mean, variance, or other texture features of the local region.

Threshold-based crack detection methods typically perform well in ideal imaging conditions, such as when cracks are clearly visible and lighting is uniform. However, real-world images are often influenced by background textures and varying lighting conditions, which introduce multiple types of noise during data acquisition. These factors make it more challenging to establish an effective threshold for distinguishing between crack and non-crack pixels. Moreover, cracks exhibit significant morphological characteristics in terms of size, topology, continuity, and contour. As a result, many approaches exploit the morphological characteristics of cracks to develop more robust detection algorithms, with edge-based methods being the most common due to the intuitive similarity between cracks and edges in terms of both topology and morphology. Filter-based methods have been widely applied to crack detection [20–23], with common spatial filters such as the Canny [24] and Sobel [25]. At the same time, to address the sensitivity of these spatial filters to geometric transformations, researchers developed crack detection methods based on frequency-domain filtering [26–29].

Filter-based methods rely on local gradient variations in the image, but random intensity fluctuations caused by noise can generate false edges, resulting in false detection. Additionally, filter-based methods are inadequate when dealing with discontinuous or morphologically complex cracks. These limitations prompted researchers to seek more robust crack detection approaches, with morphological methods and active contour models gaining significant attention. Morphological methods do not rely on intensity gradients but instead enhance or extract morphological features in the image by applying specific operations, such as dilation, erosion. These methods are commonly used in post-processing to refine edge detection results, remove noise, fill cracks, and connect discontinuities [30–32]. Active contour models iteratively evolve an initial contour in the image to adaptively approach crack edges. Since they rely on a global energy minimization process, they are better suited to handle discontinuous edges and track complex crack morphologies [33–35]. This adaptive characteristic makes them particularly effective for crack detection tasks in the presence of noise or irregular crack shapes. However, whether threshold-based or edge-based, these methods rely on predefined rules, which perform well when cracks exhibit clear, regular structures. When crack shape, size, or edge information deviates from the norm, the predefined rules may fail.

Recent research trends in crack detection have shifted towards data-driven models, which primarily rely on deep learning techniques to automatically learn segmentation rules from large-scale annotated datasets. Data-driven methods demonstrate high adaptability when handling highly complex and diverse image scenarios [3].

Fully convolutional network (FCN) [36], as one of the earliest end-to-end semantic segmentation models, convert all layers in the classification task into a fully convolutional architecture, enabling the processing of input images of arbitrary size and achieving pixel-level segmentation. This approach has shown outstanding performance in segmentation tasks across natural scenes and medical imaging, laying the foundation for subsequent research in semantic segmentation. Building on the success of FCN, many researchers quickly applied it to crack segmentation tasks [37–39], marking the first time the reliance on manually designed operators was eliminated. UNet, proposed by Ronneberger et al. for medical imaging, is one of the most widely used semantic segmentation model [40]. Its defining feature is a symmetric encoder–decoder structure, combining convolution and upsampling operations. UNet introduces skip connections for the first time, which transfer features from the encoder directly to corresponding layers in the decoder, aiding in better recovery of details. The model's sensitivity to fine details is particularly beneficial for crack segmentation tasks, as the skip connections effectively preserve spatial information lost during encoding, allowing for more precise reconstruction of crack morphology and edges during decoding [41–43]. The DeepLab series models introduced innovative techniques such as dilated convolution and atrous spatial pyramid pooling (ASPP), significantly improving the precision of multi-scale feature extraction and edge detection in semantic segmentation tasks [44,45]. These features offer notable advantages in crack segmentation tasks [46–48]. Crack segmentation not only requires accurately identifying the thin, curved edges of cracks but also capturing the overall morphology across different scales. Dilated convolution in DeepLab expands the receptive field without increasing computational costs, effectively extracting multi-scale crack features, while the ASPP module enhances the model's ability to detect cracks in complex background by fusing information from multiple scales. HRNet enhances the ability to capture fine details in segmentation tasks through its unique architecture, which maintains high-resolution feature representations throughout the network and continuously fuses features across different resolutions [49]. The model's sustained focus on high-resolution information is particularly important for crack segmentation [50]. Cracks often exhibit thin, irregular shapes, and their edges are easily obscured by complex backgrounds. Building on HRNet, many studies developed models that can precisely capture crack edges and morphology without sacrificing detail [51–53].

Through the above literature review, it is evident that the success of precise crack detection models largely depends on the accurate capture of both crack structural features and subtle details. The key to achieving this lies in the model's ability to process multi-scale features while maintaining a clear representation of fine details, addressing the complexity of crack morphology and the interference from background noise. To this end, the ECFormer model proposed in this paper adopts a dual-branch structure, ensuring the effective integration of global information while delivering high-resolution detail representation.

2.2. Transformer in crack segmentation

The Transformer model achieved groundbreaking progress in natural language processing based on its self-attention mechanism [9]. Unlike traditional sequential models, the Transformer captures long-range dependencies through self-attention while significantly improving modeling efficiency through parallel computation. In recent years, the ViT introduced the self-attention mechanism to computer vision, proposing to divide input images into fixed-size patches and process them as sequential data [10]. ViT's exceptional performance on large-scale datasets has demonstrated the potential of the self-attention mechanism in visual tasks, particularly in capturing global context and long-range feature dependencies, offering a notable advantage over CNNs in learning global features.

As Transformer models rapidly advance in vision, researchers begun exploring their potential applications in crack segmentation tasks. Drawing from the successful experiences of CNNs in crack segmentation, an increasing number of studies attempted to combine the self-attention mechanism with general CNN models to enhance the original model's ability to capture global features. For instance, several works integrated the self-attention mechanism into the UNet architecture [54–57], enabling simultaneous capture of crack details and global features through the fusion of skip connections and self-attention modules. Similarly, variants of HRNet [58,59] and DeepLab [60] leveraged the self-attention mechanism to improve performance in crack segmentation tasks, particularly in handling complex backgrounds and multi-scale crack structures. The above studies improved crack segmentation performance by introducing the self-attention mechanism, but most rely on the original softmax self-attention, which has quadratic computational complexity with respect to the input sequence length. As a result, while the addition of attention modules enhances the model's ability to capture global features, it also significantly increases the computational overhead, making the models heavier, especially when processing high-resolution images [61].

2.3. Efficient self-attention mechanism

Early improvements to the self-attention mechanism primarily focused on restricting the attention field to fixed, predefined patterns, such as local windows [62] and block patterns with fixed strides [63], thereby sparsifying the attention matrix. In contrast to fixed, predefined patterns, an extended approach involves using learnable patterns [64,65]. These learnable and efficient self-attention mechanisms dynamically adjust attention allocation in a data-driven manner, eliminating the reliance on fixed patterns. The model autonomously learns token correlations and assigns them dynamically to different buckets or clusters, optimizing both the flexibility and adaptability of the attention mechanism while reducing computational complexity. Low-rank methods and kernel methods are two major techniques developed in recent years to improve the efficiency of the self-attention mechanism. Low-rank methods aim to reduce computation and memory usage by applying low-rank approximations to the self-attention matrix. The core idea is to assume that the $N \times N$ attention matrix has a low-rank structure, allowing it to be simplified through dimensionality reduction into an $N \times k$ matrix, where k is smaller than N. Linformer is a representative of low-rank methods, achieving a significant reduction in computational complexity by projecting the length dimension of the Key and Value into a lower-dimensional representation space [66]. Kernel methods, on the other hand, improve efficiency by reformulating the attention mechanism as kernel function operations. By avoiding the explicit computation of the attention matrix, kernel methods transform attention computation into more efficient mathematical operations. Since kernel functions provide an approximation of the attention matrix, kernel methods can also be considered a variant of low-rank approaches. Relevant works include Performers [67] and Linear Transformers [68]. Another intuitive approach is to reduce the resolution of the input sequence, thereby proportionally decreasing the computational overhead. For example, Perceiver selects a small subset of Query from the input data to represent the entire sequence [69], while Funnel Transformer progressively reduces the sequence length layer by layer, following a pyramid-like structure, which gradually lessens the computational costs at each layer [70].

3. Preliminaries

3.1. Self-attention mechanism

Self-attention operates by calculating a set of attention weights for each element in the input sequence, representing the relative importance of other elements in relation to the current element [9]. In this way, self-attention enables the model to dynamically adjust its focus based on context, capturing long-range dependencies that traditional methods, such as convolutional or recurrent layers, may overlook. The workflow of the self-attention mechanism involves the following steps:

1. Projection of Query, Key, and Value. For each element x_i in the input sequence $X \in \mathbb{R}^{N\times d_{\text{in}}}$, linear transformations are applied to project it into a query vector Q_i, a key vector K_i, a value vector V_i. The process is formulated as:
$$Q_i = W_Q \cdot x_i, \quad K_i = W_K \cdot x_i, \quad V_i = W_V \cdot x_i \tag{1}$$
where W_Q, W_K and W_V are learnable projection matrices for Query, Key and Value, respectively.
2. Computation of attention weights. Each Q_i is multiplied by all K_i using dot product to compute the attention weights, which are then normalized through the softmax function. The formula is as follows:
$$a_{ij} = \frac{\exp\left(Q_i \cdot K_j\right)}{\sum_{j=1}^{N} \exp\left(Q_i \cdot K_j\right)} \tag{2}$$
3. Weighted sum. The final output is obtained by performing a weighted sum of the value vectors V_j, where the weights are the computed attention weights a_{ij}. The formula for calculating the output O_i is:
$$O_i = \sum_{j=1}^{N} a_{ij} \cdot V_j \tag{3}$$
The weighted summation enables the model to emphasize the most important parts of the input sequence relative to the current element x_i, which is the essence of self-attention.

When calculating the attention weights, each Q_i must perform a dot product operation with all K_j, resulting in a computational complexity of $\mathcal{O}(N^2)$, where N represents the length of the input sequence.

3.2. Self-attention in vision

The ViT is a representative model of self-attention in vision [10]. Its core idea is to divide the input image into fixed-size patches and treat these patches as sequence elements, then apply the attention mechanism to capture global features of the image.

1. Patch embedding. First, the input image $X \in \mathbb{R}^{H\times W\times C}$ is divided into N non-overlapping patches, each of size $P \times P$. The patches are flattened into vectors $x_i \in \mathbb{R}^{P^2\cdot C}$. Then, each patch is projected into another space through a linear projection, yielding the following embedding representation:
$$z_i = E \cdot x_i + e_{\text{pos}} \tag{4}$$
Where, $E \in \mathbb{R}^{D\times P^2\cdot C}$ is the projection matrix, and e_{pos} represents the positional embedding, which retains the spatial information of the patches within the original image. The sequence of embeddings is then represented as $Z = [z_1, z_2, \dots, z_N]$.
2. ViT Block. In ViT, the input sequence Z is processed through multiple Transformer blocks. Each ViT block consists of Multi-head Self-Attention (MSA) and Layer Normalization (LN) and a Feed-Forward Network (FFN). The computation flow for each block can be expressed as:
$$\begin{aligned} Z' &= \text{MSA}(\text{LN}(Z)) + Z \\ Z'' &= \text{FFN}(\text{LN}(Z')) + Z' \end{aligned} \tag{5}$$
In the first step, the input sequence Z is passed through LN and then fed into MSA. The output is then added back to the original input Z via a residual connection. In the second step, the intermediate result Z' is again normalized and passed through the FFN, where non-linear transformations are applied. The result is then added back to Z' through another residual connection, yielding the final output Z''.

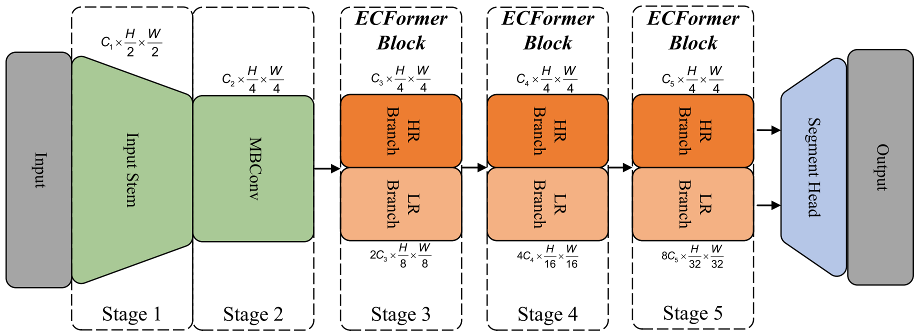


Fig. 2. Overall architecture of ECFormer.

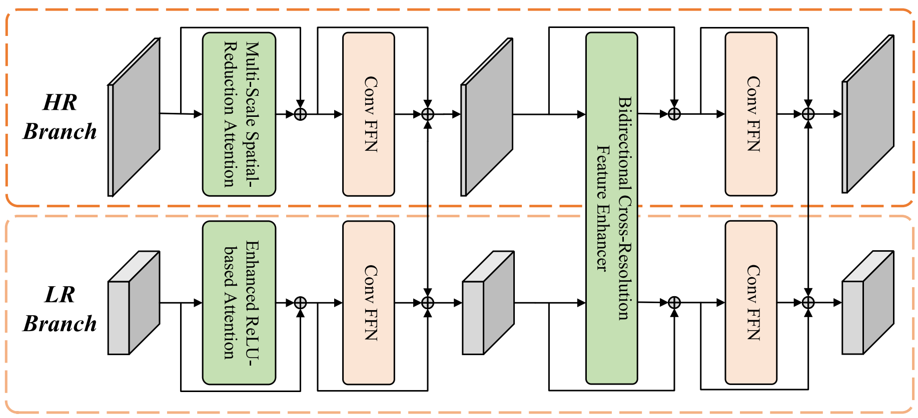


Fig. 3. ECFormer block.

4. Methodology

In this section, we describe the overall model architecture, followed by a detailed introduction to its three key components: Multi-Scale Spatial-Reduction Attention (MSSRA), Enhanced ReLU-based Linear Attention (ERLA), and Bidirectional Cross-Resolution Feature Enhancer (BDCRFE).

4.1. Overall architecture

Fig. 2 illustrates the overall structure of ECFormer, a five-stage model that integrates several key components to ensure efficient and accurate crack segmentation. The first stage is the input stem, consisting of standard convolution, batch normalization (BN), and ReLU, which reduces the spatial resolution of the input image by half while expanding the channel dimension to C_1. In the second stage, the resolution is further reduced to 1/4 of the original image by the MBConv blocks [71], while the channel dimension increases to C_2. The use of MBConv is intended to reduce computational costs. The third to fifth stages form the core of the model, consisting of ECFormer blocks, which serve as the backbone of the entire architecture. Each ECFormer block contains two branches, a high-resolution (HR) branch and a low-resolution (LR) branch. The design of resolution variation reflects the different functions of these two branches. The HR branch maintains a fixed resolution of $H/4 \times W/4$ throughout the model to ensure that critical detail information is preserved during feature extraction, thereby enhancing the model's ability to find small cracks. In contrast, the LR branch follows a stepwise downsampling strategy, with the resolution decreasing from $H/8 \times W/8$ in the third stage to $H/16 \times W/16$ in the fourth stage, and finally to $H/32 \times W/32$ in the fifth stage. The progressive reduction in resolution allows the model to extract high-level semantic features while maintaining an effective understanding of global crack structures.

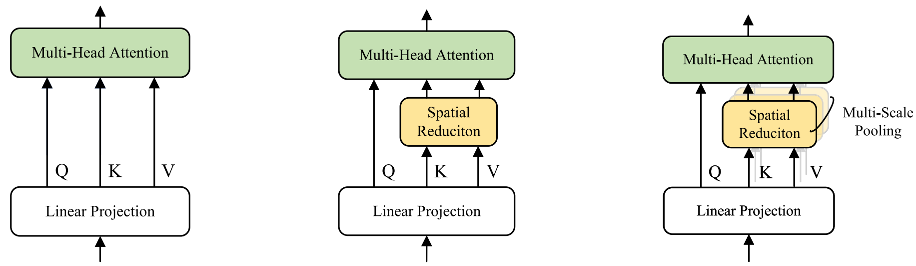
The HR and LR branches work in parallel, with their outputs fused at each stage to achieve a rich multi-scale representation. The dual-branch structure effectively preserves multi-scale learning and high-resolution representation, which are critical for accurate crack segmentation.

4.2. ECFormer block

The ECFormer blocks form the core of the proposed model, featuring not only an efficient dual-branch design but also three crafted modules: MSSRA, ERLA, and BDCRFE, as shown in Fig. 3.

4.2.1. Multi-scale spatial-reduction attention

The original MSA [9] takes an input $X \in \mathbb{R}^{N\times d_{\text{in}}}$, and then applies a linear projection layer to transform the input X into $Q, K, V \in \mathbb{R}^{N\times d}$. The self-attention then takes Q, K, and V as inputs and outputs new middle features. As analyzed in Section 3.1, the primary issue with the self-attention in terms of efficiency is that its computational complexity relative to the input sequence is $\mathcal{O}(N^2)$. Wang et al. [72,73] proposed spatial-reduction attention (SRA) to improve the efficiency of the original self-attention. They introduced an additional spatial-reduction module between the linear projection layer and the MSA, as shown in Fig. 4(b). Specifically, the spatial reduction module first performs downsampling on the spatial dimension of K and V, transforming the high-resolution feature map into a low-resolution. The low-resolution feature map is then processed by the MSA block. Let the downsampling factor be r, and the sequence length of the input after



(a) Original self-attention (b) Spatial-reduction attention (c) Multi-scale spatial-reduction attention

Fig. 4. Comparison of original self-attention, spatial-reduction attention, multi-scale spatial-reduction attention.

downsampling is N/r. Therefore, the computational complexity of SRA is $\mathcal{O}((N/r)^2 \times d)$, which can be simplified to $\mathcal{O}((N^2 \times d)/r^2)$. Since r is usually greater than 1, which means that the computational complexity of SRA is significantly lower than that of the original self-attention. For the input X, SRA can be expressed as:

$$\mathrm{SRA}(X) = \mathrm{MSA}(Q, \mathrm{SR}(K), \mathrm{SR}(V)) \tag{6}$$

There are typically two methods for spatial reduction: one is convolution with a stride greater than 1, and the other is pooling. Since pooling has significantly lower complexity than convolution and does not involve learnable parameters, this study adopts adaptive average pooling following PVTv2 [73].

SRA reduces computational complexity compared to the original self-attention but limits the model's capacity. Since the spatial-reduction module downsamples K and V to a fixed spatial resolution, the Q can only retrieve features at a fixed scale. For example, in PVTv2 [73], the authors fixed the reduced spatial resolution to 7×7. To enhance the ability to learn multi-scale spatial features, our paper proposes a multi-scale pooling to capture features at different scales, referred to as multi-scale spatial-reduction attention, as shown in Fig. 4(c). Unlike window-based approaches such as Swin Transformer [62], which restrict attention to local regions, MSSRA maintains a global receptive field while capturing features at multiple scales. This is particularly advantageous for segmenting cracks, which often exhibit long-range, continuous structures that would span across local windows.

MSSRA includes four different global adaptive average pooling sizes, specifically 1, 3, 5, and 7. In the implementation, after linear projection, Q, K, and V are divided into four parts along the channel dimension, following the multi-head setting of the original self-attention. In this work, the number of heads is kept at four. Spatial pooling is applied to K and V in each head, resulting in four new sets of K and V at different scales. Self-attention is then performed on the feature maps at different scales, producing multi-scale outputs. Finally, the multi-scale features are concatenated along the channel dimension. The proposed MSSRA can be expressed as:

$$\begin{aligned} \mathrm{MSSRA} &= \mathrm{Linear}(\mathrm{Concat}(\mathrm{Head}_1, \ldots, \mathrm{Head}_n)) \\ \mathrm{Head}_i &= \mathrm{Attention}(Q_i, \mathrm{SR}(K_i), \mathrm{SR}(V_i)) \end{aligned} \tag{7}$$

where i represents different attention heads, which are set to 4 in this work. Each head performs adaptive global average pooling with one of the sizes from 1, 3, 5, and 7, meaning that each head extracts spatial features at a fixed scale.

The reduction in computational complexity brought by MSSRA is particularly beneficial for handling high-resolution features, which typically have very long sequences. Directly applying the self-attention would result in significant computational and memory overhead. By introducing the multi-scale spatial-reduction module, the demand for computational resources can be greatly reduced while maintaining a certain level of feature representation capability. Specifically, given input features of size $H \times W \times C$, the computational costs of MSA, SRA, and MSSRA can be expressed as:

$$\begin{aligned} \Omega(\mathrm{MSA}) &= 2(HW)^2 C \\ \Omega(\mathrm{SRA}) &= 2HWp^2 C \\ \Omega(\mathrm{MSSRA}) &= \sum_{i=1}^{4} 2HWp_i^2 \frac{C}{4} \\ &= \frac{1}{2} \sum_{i=1}^{4} HWp_i^2 C \end{aligned} \tag{8}$$

Here, p represents the size of the adaptive pooling. When p is set to 7, the computational complexity of the proposed MSSRA is only half that of the original SRA [73]. Given its efficiency on high-resolution features, this study applies the proposed MSSRA to the HR branch of ECFormer.

4.2.2. Enhanced ReLU-based linear attention

In crack segmentation, capturing global information is crucial for improving model performance. Although the original self-attention provides a global receptive field, it comes with high computational complexity. To address this issue, this work adopts a ReLU-based linear attention (RLA) [68].

Given input $X \in \mathbb{R}^{N \times d_{\mathrm{in}}}$, the general form of self-attention can be represented as:

$$A_i = \sum_{j=1}^{N} \frac{\mathrm{Sim}\left(Q_i, K_j\right) V_j}{\sum_{j=1}^{N} \mathrm{Sim}\left(Q_i, K_j\right)} \tag{9}$$

Where $Q = XW_Q$, $K = XW_K$, $V = XW_V$, and $W_Q, W_K, W_V \in \mathbb{R}^{d_{\mathrm{in}} \times d}$ are the learnable linear projection matrices. A_i indicates the i-th row of attention matrix A. $\mathrm{Sim}(\cdot,\cdot)$ denotes a function that calculates the similarity between two vectors. In the original self-attention [9] and ViT [10], the commonly employed method is scaled dot-product attention with softmax, where the similarity metric function Sim(.,.) is:

$$\mathrm{Sim}(Q, K) = \exp\left(\frac{QK^{\mathrm{T}}}{\sqrt{d}}\right) \tag{10}$$

where the attention matrix is obtained by calculating the similarity between all Q and K, leading to a computational complexity of $\mathcal{O}(N^2)$. Apart from the similarity metric function in Eq. (10), Angelos et al. [68] proposed linear attention as an alternative to softmax attention, which is generally represented as:

$$\mathrm{Sim}(Q, K) = \phi(Q)\phi(K)^T \tag{11}$$

where, $\phi()$ is a kernel function. Based on Eq. (11), Eq. (9) can be rewritten as:

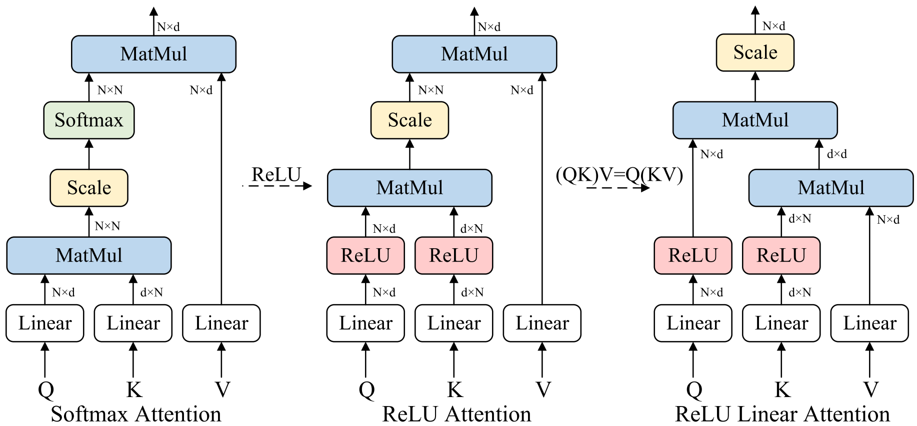


Fig. 5. Computational flow of different attention.

$$A_i = \sum_{j=1}^{N} \frac{\phi\left(Q_i\right)\phi\left(K_j\right)^{\mathrm{T}} V_j}{\sum_{j=1}^{N}\phi\left(Q_i\right)\phi\left(K_j\right)^{\mathrm{T}}} = \frac{\sum_{j=1}^{N}\left(\phi(Q_i)\phi(K_j)^{\mathrm{T}}\right)V_j}{\phi(Q_i)\sum_{j=1}^{N}\phi(K_j)^{\mathrm{T}}} \tag{12}$$

According to the associative property of matrix multiplication, Eq. (12) can be further optimized as:

$$A_i = \frac{\sum_{j=1}^{N}\left[\phi\left(Q_i\right)\phi\left(K_j\right)^{\mathrm{T}}\right]V_j}{\phi\left(Q_i\right)\sum_{j=1}^{N}\phi\left(K_j\right)^{\mathrm{T}}} = \frac{\sum_{j=1}^{N}\phi\left(Q_i\right)\left[\left(\phi\left(K_j\right)^{\mathrm{T}}V_j\right)\right]}{\phi\left(Q_i\right)\sum_{j=1}^{N}\phi\left(K_j\right)^{\mathrm{T}}} = \frac{\phi\left(Q_i\right)\left(\sum_{j=1}^{N}\phi\left(K_j\right)^{\mathrm{T}}V_j\right)}{\phi\left(Q_i\right)\left(\sum_{j=1}^{N}\phi\left(K_j\right)^{\mathrm{T}}\right)} \tag{13}$$

It can be observed that when calculating the attention matrix A, it is only necessary to compute $\sum_{j=1}^{N}\phi(K_j)^{\mathrm{T}}V_j$ and $\sum_{j=1}^{N}\phi(K_j)^{\mathrm{T}}$ once, which can then be reused for each Q. Therefore, the computational complexity is $\mathcal{O}(N)$, meaning linear complexity. Researchers in EfficientViT employed ReLU as the kernel function and demonstrated its effectiveness in the visual domain [61]. Fig. 5 presents a comparison of the computational flow between softmax attention and ReLU-based linear attention.

Compared to the original softmax attention [9,10], ReLU-based linear attention [68] offers significant advantages in terms of computational complexity and hardware efficiency. However, it still has some drawbacks. Specifically, softmax possesses a non-linear normalization property, where all input vectors are normalized into a probability distribution within the range [0,1]. Additionally, since the sum of all elements equals 1, which ensures a competitive relationship among the elements, making the attention map more focused and thus better at capturing more relevant information. In contrast, the ReLU function truncates negative values in the input vectors to 0, resulting in sparse feature maps. Due to the lack of normalization and non-linear competition, the feature maps generated by ReLU tend to be smoother, making it harder to highlight important attention regions and less effective at capturing key features compared to softmax attention. Based on the above analysis, this work proposes an enhancement to ReLU-based linear attention by applying an exponential function mapping to the input tensor. By adjusting the direction of each Q and K, it brings closer the Q/K pairs that are more similar and pushes apart those that are less similar. The proposed new similarity metric function:

$$\mathrm{Sim}\left(Q_i, K_j\right) = \phi_p\left(Q_i\right)\phi_p\left(K_j\right)^{\mathrm{T}} \tag{14}$$

Where,

$$\phi_p(x) = f_p(\mathrm{ReLU}(x)) \qquad f_p(x) = \frac{\|x\|}{\|e^x - 1\|}\left(e^x - 1\right) \tag{15}$$

To provide a more intuitive understanding, an example is given here to demonstrate the effect of the mapping function f_p, as shown in Fig. 6. It can be directly observed that the norm of the feature vectors remains unchanged before and after the f_p mapping, while only the direction changes. At the same time, features that are close to each other are drawn closer, whereas those that are farther apart become more distant. This ensures that the distribution of the feature matrix becomes sharper, allowing the model to focus on the most important features. While other linear attention methods like Performer [67] achieve efficiency by approximating the softmax function with feature kernels, the unique contribution of ERLA is this explicit feature sharpening via the f_p function.

The LR branch has the ability to learn global semantic features [50,74], while the global receptive field can capture the overall structure of the image and long-range dependencies [44]. The proposed ERLA module achieves a global receptive field with only linear complexity. Therefore, this paper applies the module to the LR branch.

4.2.3. Bidirectional cross-resolution feature enhancer

In most dual-branch or multi-branch models [49,50,74], the feature fusion from different branches is often based on element-wise multiplication or addition, which are highly efficient methods. However, such simple fusion techniques may not fully capture the complex relationships between different branches. To better utilize the features extracted by each branch, several recent works have proposed more complex fusion methods. For instance, attention-based fusion methods dynamically assign weights to different features, allowing for a better emphasis on key features and the suppression of irrelevant information [75,76]. Attention-based feature enhancement methods are particularly advantageous in vision-language multi-modal tasks [77,78], as these models need to handle different types of features. Cross-attention can facilitate interactive modeling between different modalities, capturing correlations across modalities and improving performance [9]. The motivation for the proposed BDCRFE in this work stems from this idea, leveraging the well-established cross-attention mechanism that has been proven effective in multi-modal learning and multi-scale vision transformers.

Specifically, the proposed method considers the HR and LR branches as two different modalities and uses bidirectional cross-resolution cross-attention to enhance feature fusion. The work most similar to the

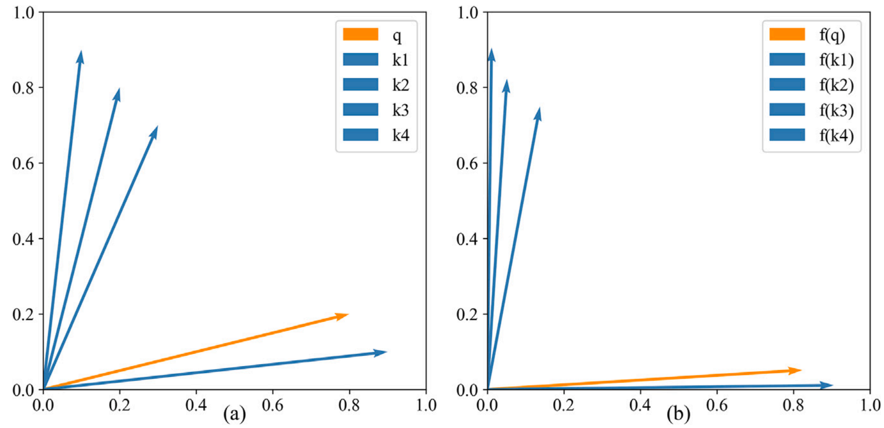


Fig. 6. Mapping function f_p on Q/K.

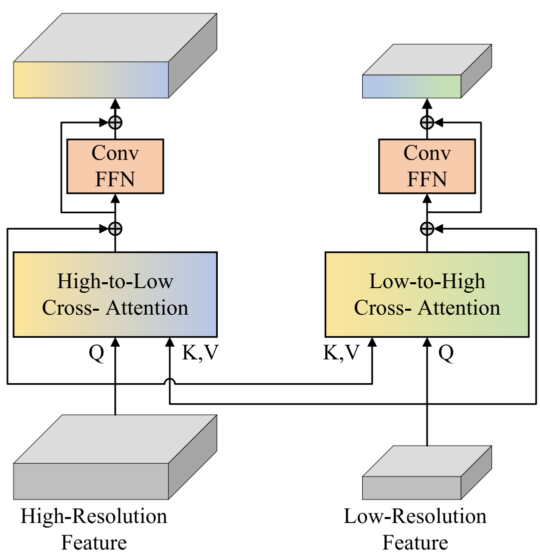


Fig. 7. Bidirectional cross-resolution feature enhancer.

Algorithm 1 Bidirectional cross-resolution feature enhancer.

Input: High-res feature F_{HR}, Low-res feature F_{LR}
Output: Enhanced features $F'_{\mathrm{HR}}, F'_{\mathrm{LR}}$
#— High-to-Low Cross-Attention (H2LCA) —
$Q_{\mathrm{HR}} \leftarrow F_{\mathrm{HR}}$
$K_{\mathrm{LR}}, V_{\mathrm{LR}} \leftarrow F_{\mathrm{LR}}, F_{\mathrm{LR}}$
$K'_{\mathrm{LR}} \leftarrow \mathrm{Conv}_{1\times1}^{\mathrm{LR}\to\mathrm{HR}}(K_{\mathrm{LR}})$ ▷ Align K channels to HR
$V'_{\mathrm{LR}} \leftarrow \mathrm{Conv}_{1\times1}^{\mathrm{LR}\to\mathrm{HR}}(V_{\mathrm{LR}})$ ▷ Align V channels to HR
$F'_{\mathrm{HR}} \leftarrow \mathrm{MSSRA}(Q_{\mathrm{HR}}, K'_{\mathrm{LR}}, V'_{\mathrm{LR}}) + F_{\mathrm{HR}}$ ▷ Residual connection
#— Low-to-High Cross-Attention (L2HCA) —
$Q_{\mathrm{LR}} \leftarrow F_{\mathrm{LR}}$
$K_{\mathrm{HR}}, V_{\mathrm{HR}} \leftarrow F_{\mathrm{HR}}, F_{\mathrm{HR}}$
$K'_{\mathrm{HR}} \leftarrow \mathrm{Conv}_{1\times1}^{\mathrm{HR}\to\mathrm{LR}}(K_{\mathrm{HR}})$ ▷ Align K channels to LR
$V'_{\mathrm{HR}} \leftarrow \mathrm{Conv}_{1\times1}^{\mathrm{HR}\to\mathrm{LR}}(V_{\mathrm{HR}})$ ▷ Align V channels to LR
$F'_{\mathrm{LR}} \leftarrow \mathrm{ERLA}(Q_{\mathrm{LR}}, K'_{\mathrm{HR}}, V'_{\mathrm{HR}}) + F_{\mathrm{LR}}$ ▷ Residual connection
return $F'_{\mathrm{HR}}, F'_{\mathrm{LR}}$

proposed method is CrossViT [75], where the authors divide the same image into two groups of patches with different sizes and apply self-attention within each group before using cross-attention to fuse features. However, the design of CrossViT is aimed at image classification, and the authors only used the CLS token as a Query during the cross-attention fusion stage to exchange information between the two branches. While this design is sufficient for image classification, it is inadequate for dense prediction tasks.

The BDCRFE proposed in this work is more generalized, fully exchanging features from both branches. Additionally, through stacking multiple ECFormer modules, the proposed feature enhancer endows the model with multi-scale learning capabilities on a global level. Specifically, in the three ECFormer blocks, the scale of the HR branch remains constant, while the spatial resolution of the LR branch progressively decreases, ensuring that the HR branch can learn features of different scales from the LR branch.

Fig. 7 illustrates the proposed BDCRFE. The HR features are derived from the previous MSSRA module, while the LR features come from the preceding ERLA module. To establish an effective connection between the HR and LR features, the BDCRFE module achieves feature fusion in two steps. First, the proposed method utilizes high-to-low cross attention (H2LCA), where the HR features are used as the Query, and the LR features serve as the Key and Value, facilitating feature interaction and fusion. This design builds connections between features of different resolutions, allowing the HR features to capture the global information from the LR features, thereby enhancing the HR representational capacity. Next, through low-to-high cross-attention (L2HCA), the LR features are used as the Query, and the HR features as the Key and Value, performing feature interaction and fusion once again. This step complements the previous H2LCA, enabling the LR features to absorb detailed information from the HR branch.

Notably, for the bidirectional cross-attention in Fig. 7, the channel numbers of the Key and Value differ from those of the Query, so a 1×1 convolution is used to align the channel dimensions. Additionally, the MSSRA and ERLA proposed in the Sections 4.2.1 and 4.2.2 are used to replace the original cross-attention in H2LCA and L2HCA, and a detailed comparison is provided in the ablation analysis. The detailed steps of the BDCRFE are presented in Algorithm 1.

4.3. Model instantiation

Following HRNet [49] and HRSegNet [50], the channel dimension in the HR branch of ECFormer is fixed at a constant value. In this study, three variants of ECFormer, namely ECFormer-B16, ECFormer-B32, and ECFormer-B48, are employed for experiments, where 16, 32, and 48 represent the channel dimensions of the HR branch.

Table 1 provides details of the ECFormer configuration. ECFormer consists of two branches, where the HR branch remains throughout the entire model, while the LR branch only exists within the ECFormer block.

Table 1
Detailed configuration of ECFormer. s denotes convolution stride, t represents the intermediate expansion rate of MBConv, p refers to the maximum adaptive pooling size of MSSRA, and h indicates the number of heads for all self-attention.

Stage	Module	HR Branch	LR Branch
1	Input Stem	Conv-BN-ReLU, $s = 1$ Conv-BN-ReLU, $s = 2$	–
2	MBConv	MBConv, $s = 1, t = 4$ MBConv, $s = 2, t = 4$	–
3	ECFormer Block	MSSRA, $p = 7, h = 4$ BDCRFE, $h = 4$	ERLA, $h = 4$ BDCRFE, $h = 4$
4	ECFormer Block	MSSRA, $p = 7, h = 4$ BDCRFE, $h = 4$	ERLA, $h = 4$ BDCRFE, $h = 4$
5	ECFormer Block	MSSRA, $p = 7, h = 4$ BDCRFE, $h = 4$	ERLA, $h = 4$ BDCRFE, $h = 4$

Table 2
Details of datasets.

Dataset	Size	Training Set	Validation Set	Test Set
CrackSeg9k	400×400	6951	900	900
Concrete3k	500×500	1800	300	900
Asphalt3k	500×500	1800	300	900

The channel dimension of the LR branch doubles as the spatial resolution is halved, following the design of ResNet [79]. For the model's segmentation head, a dual-step segmentation head from HRSegNet is used [50].

In recent Transformer-based segmentation models, the FFN has gradually shifted from a two-layer MLP [10] to depth-wise convolution [73,76]. This is because convolution has been shown to complement positional information and locality, thereby eliminating the need for positional encoding. This paper follows the design of RTFormer [76], using two 3×3 depth-wise convolution layers without dimensional expansion in the attention modules of MSSRA, ERLA, and BDCRFE. This setup uses fewer parameters.

5. Experiments

5.1. Datasets and evaluation metrics

In this study, three datasets, CrackSeg9k [80], Concrete3k [50], and Asphalt3k [50] are used to validate the proposed model. CrackSeg9k is currently a relatively large publicly available evaluation benchmark in the field of crack segmentation, which incorporates ten public crack datasets containing four different materials, namely asphalt, concrete, masonry, and ceramic. CrackSeg9k contains a total of 8751 images with a resolution of 400×400 and their corresponding labels, which are assigned to the categories of crack and background. It should be noted that since the original dataset suffers from data and label mismatches, the version in the literature HrSegNet [50] is used in this work. This paper randomly selects 900 images as the validation and test sets, respectively, and the rest are used as the training set. Concrete3k and Asphalt3k are scenario-specific crack datasets focusing on the task of detecting cracks in concrete and asphalt structures respectively, and both of them have a resolution of 500×500. Both datasets are divided into training, validation, and test sets according to the ratio of 6:1:3. The details of the three datasets are in Table 2 and some samples are shown in Fig. 8.

Four metrics are used in this work to evaluate the segmentation performance of all models, Precision (Pr), Recall (Re), F1-score and mean Intersection over Union (mIoU). These metrics are defined as follows:

$$\mathrm{Pr} = \frac{TP}{TP + FP} \tag{16}$$

$$\mathrm{Re} = \frac{TP}{TP + FN} \tag{17}$$

$$\mathrm{F1} = 2 \times \frac{\mathrm{Pr} \times \mathrm{Re}}{\mathrm{Pr} + \mathrm{Re}} \tag{18}$$

$$\mathrm{IoU} = \frac{TP}{TP + FP + FN} \tag{19}$$

$$\mathrm{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{IoU}_i \tag{20}$$

Where True Positive (TP) means that crack pixels are correctly classified, False Positive (FP) means that background pixels are incorrectly classified as cracks, and False Negative (FN) means that crack pixels are incorrectly recognized as background.

In addition, Frames Per Second (FPS) is used to evaluate the inference speed, and Giga Floating-Point Operations (GFLOPs) and Parameters (Params) are used as metrics to evaluate the computational complexity and the model size.

5.2. Implementation details

The proposed ECFormer instances are implemented using PyTorch [81], with all training, evaluation, and testing conducted within the MMSegmentation framework [82]. During training, data augmentation including random resizing, random horizontal flipping, random cropping, and random photometric distortion are applied. The random resizing scale ranges from 0.5 to 2.0, with a crop size of 400×400 for CrackSeg9k, and 500×500 for both Concrete3k and Asphalt3k. The ECFormer instances are trained using the AdamW [83], with 100,000 iterations on CrackSeg9k and 20,000 iterations on Concrete3k and Asphalt3k. Notably, for the ablation studies on CrackSeg9k, the random crop size is 200×200, with 20,000 iterations per experiment to quickly validate the impact of each module on model performance. The initial learning rate is set to 0.00006, with a default polynomial learning rate scheduler using a factor of 1.0, and weight decay set to 0.01. All training experiments for ECFormer, including the measurements for training time and GPU memory, are conducted on a single NVIDIA RTX 4090 with a batch size of 16. The inference-speed evaluations are performed on an RTX 4060 with a batch size of 1. All models use a binary cross-entropy loss.

5.3. Comparative study and results

In evaluating ECFormer models, their performance is compared with commonly used and state-of-the-art deep learning methods. The comparison models are broadly divided into two categories: CNN-based and Transformer-based models. The CNN models include FCN [36], UNet [40], BiSeNetV2 [84], OCRNet [85], and DMA-Net [47], while the Transformer-based models are SegFormer [86], RTFormer [76], CrackFormer [54], CT [87], and SegCrack [88]. Among them, DMA-Net, CrackFormer, CT, and SegCrack represent leading models in crack segmentation since 2022. Table 3 presents the quantitative evaluation results of all models on CrackSeg9k.

5.3.1. Quantitative analysis

Results from Table 3 indicate that ECFormer excels across multiple key performance metrics. Specifically, ECFormer-B48 leads with the highest mIoU at 81.85 %, indicating it provides the most accurate segmentation results compared to all other models. Additionally, ECFormer-B48 exhibits exceptional performance in Pr and Re, with scores of 89.04 % and 87.54 % respectively, and an F1 score of 88.28 %. Compared to CNN-based models like FCN, UNet, and DMA-Net, the ECFormer series show a clear advantage. Although OCRNet achieves an mIoU of 80.90 %, ranking second to ECFormer-B48, its parameters (12.11 M) and computational complexity (32.40 GFLOPs) are significantly higher than those of ECFormer-B48 (6.08 M and 6.54 GFLOPs), indicating that ECFormer maintains high performance while being more computationally and parameter efficient. Among Transformer-based models, ECFormer variants also demonstrate significant superiority. For instance, SegFormer's mIoU of 78.80 % is considerably lower

(a) CrackSeg9k (b) Concrete3k (c) Asphalt3k

Fig. 8. Samples of CrackSeg9k, Concrete3k, Asphalt3k.

Table 3
Quantitative evaluation on CrackSeg9k.

Model	mIoU (%)	Pr (%)	Re (%)	F1 (%)	FPS	Params (M)	GFLOPs	Memory(GB)	Train Time(h)
FCN (MobileNetV2)	77.84	87.22	81.25	84.13	66.2	9.80	29.60	6.3	6.1
UNet	79.15	89.82	84.75	87.21	45.7	13.40	75.87	19.4	15.7
BiSeNetV2	75.09	87.07	81.17	84.38	85.3	2.33	4.93	5.8	2.1
OCRNet (HRNet-W18)	80.90	88.26	88.58	88.41	45.2	12.11	32.40	14.6	6.7
DMA-Net (ResNet18)	80.67	88.95	87.23	88.08	58.8	13.80	44.26	10.4	9.2
SegFormer (MiT-B0)	78.80	88.99	83.56	86.19	95.5	3.81	6.65	6.7	2.9
RTFormer-Slim	79.96	89.05	86.34	87.67	75.1	4.80	3.94	5.5	2.3
CrackFormer	80.25	88.04	86.61	87.32	27.9	4.96	59.21	19.6	12.3
CT	79.27	90.21	85.33	87.70	68.2	6.62	17.62	11.9	4.0
SegCrack	79.59	88.10	85.27	86.66	35.5	28.44	42.83	20.5	9.8
ECFormer-B16	80.84	88.74	87.12	87.92	156.2	0.82	0.98	3.6	1.1
ECFormer-B32	81.28	88.91	87.03	87.96	132.3	2.89	2.83	4.9	1.9
ECFormer-B48	81.85	89.04	87.54	88.28	102.9	6.08	6.54	6.1	2.6

than that of ECFormer-B48. Furthermore, although RTFormer-Slim and CrackFormer have advantages in terms of parameters, their overall performance still falls short of ECFormer. Particularly, CrackFormer, with an mIoU of 80.25 %, remains below ECFormer-B48's 81.85 %.

Importantly, ECFormer demonstrates a significant advantage in inference speed. ECFormer-B16 achieves an inference speed of 156.2 FPS, substantially higher than other models. Even with increased parameters and computational demand, ECFormer-B32 and ECFormer-B48 maintain high real-time performance, with 132.3 and 102.9 FPS, respectively.

ECFormer demonstrates excellent scalability through its variants B16, B32, and B48, enabling efficient performance across diverse computational resources. A model's scalability is primarily characterized by its consistent performance and efficiency optimization at different scales. The B16, B32, and B48 variants represent different configurations of channel dimensions in the HR branch. Specifically, ECFormer-B16 has the smallest parameters (0.82 M) and computational complexity (0.98 GFLOPs), while ECFormer-B48 has the largest (6.08 M parameters, 6.54 GFLOPs). This design allows ECFormer to adapt flexibly to a range of computational devices and application scenarios, from low-end to high-end. Table 3 indicates that despite significant differences in parameters and computational complexity, ECFormer variants maintain high consistency in performance metrics. For instance, ECFormer-B16 achieves an mIoU of 80.84 %, Pr of 88.74 %, Re of 87.12 %, and F1 score of 87.92 %. As model size increases, ECFormer-B32 and ECFormer-B48 show improvements in these metrics, with B48 reaching the highest mIoU of 81.85 %, Pr of 89.04 %, Re of 87.54 %, and F1 score of 88.28 %. The observed steady improvement in model performance as scale increases demonstrates ECFormer's scalability advantage.

5.3.2. Qualitative analysis

Figs. 9 and 10 visually present some results of ECFormer and other methods on the CrackSeg9k test set. Overall, Transformer-based models outperform some CNN-based models, particularly FCN and BiSeNetV2, which consistently produce weaker results compared to Transformer-based models. This highlights the advantage of Transformer models in capturing long-range dependencies and complex structures in crack images, leading to more accurate segmentation and validating the method selection in this work. While OCRNet is also a CNN-based model, its performance surpasses the other four CNN models, largely due to its HRNet backbone [49], further supporting the rationale for designing high-resolution models in our paper. For ECFormer specifically, as the model scale increases from B16 to B48, segmentation performance steadily improves, demonstrating the stable scalability of the proposed approach.

The qualitative results presented in Figs. 9 and 10 not only highlight ECFormer's superior segmentation accuracy but also provide compelling evidence of its robustness across various challenging real-world scenarios. For instance, in the case of thin, evenly distributed cracks (first row), Transformer-based models like ECFormer accurately capture these fine details, whereas traditional CNN models exhibit significant missed detections. For cracks captured from an oblique viewpoint with complex morphology (second row), ECFormer achieves the most accurate and complete segmentation compared to the rough results from other models. When faced with low-light and low-contrast images (rows three and five), ECFormer-B48 successfully identifies the entire crack outline, a task where other models, including FCN, UNet, and SegCrack, substantially fail. Furthermore, for images featuring complex background textures and noise (last row), ECFormer demonstrates strong noise resilience by accurately segmenting the cracks and avoiding false positives.

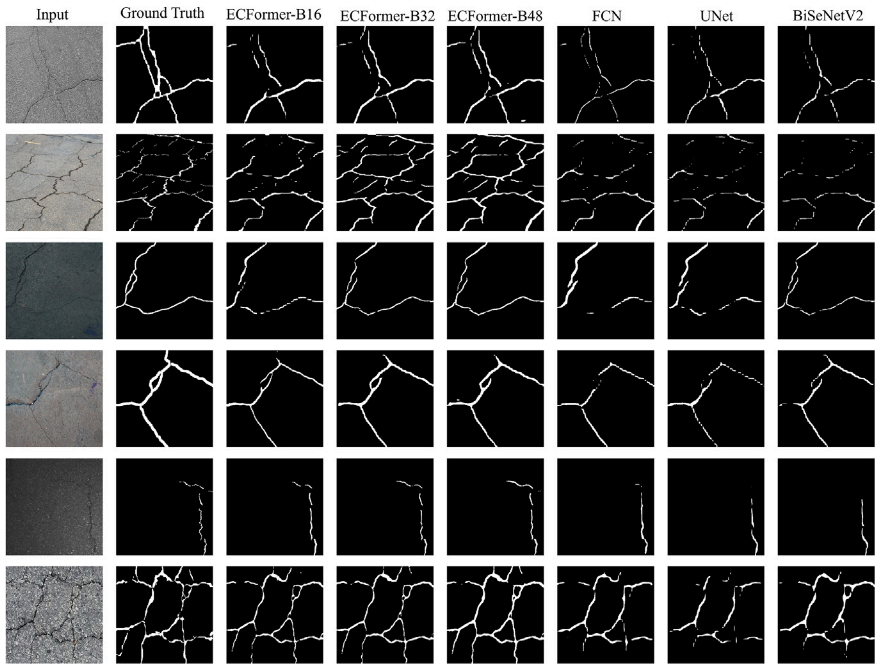


Fig. 9. Qualitative results of ECFormer, FCN, UNet, and BiSeNetV2 on CrackSeg9k.

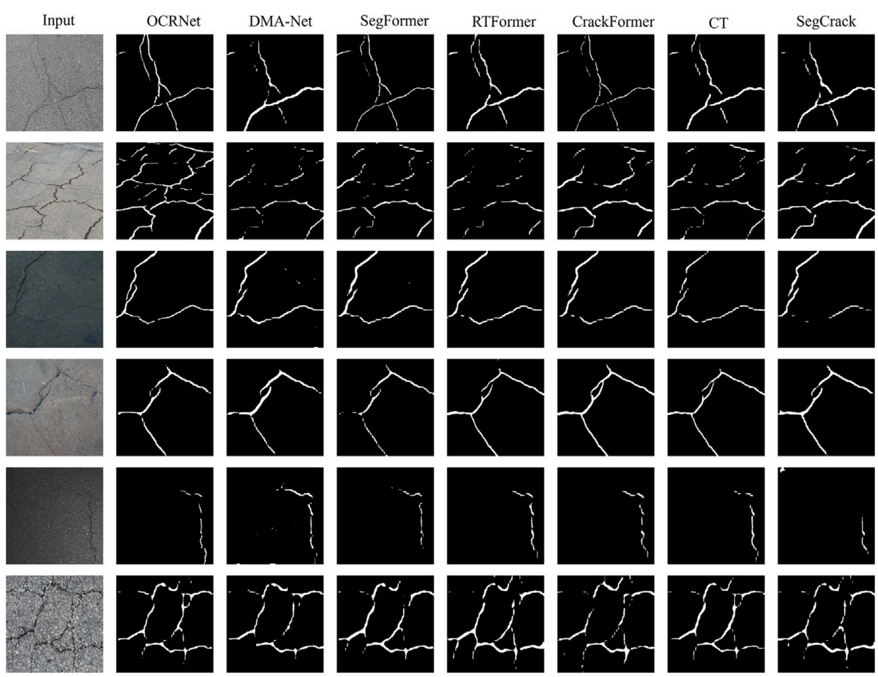


Fig. 10. Qualitative results of OCRNet, DMA-Net, SegFormer, RTFormer, CrackFormer, CT and SegCrack on CrackSeg9k.

In summary, ECFormer excels in diverse and challenging conditions—from tiny cracks and complex viewpoints to low contrast and noisy backgrounds. This consistent outperformance against both CNN and other Transformer models validates its superiority and confirms the robust design of the proposed architecture for crack segmentation tasks.

5.3.3. Results in specific scenarios

In practical applications, crack detection tasks typically involve specific scenarios rather than a comprehensive dataset like CrackSeg9k. Therefore, the performance of ECFormer on Concrete3k and Asphalt3k is analyzed to assess its effectiveness in specific contexts. Table 4 compares the crack segmentation performance of models that performed well on CrackSeg9k with their results on Concrete3k and Asphalt3k, while Figs. 11 and 12 provide visual representations.

Table 4 shows that ECFormer performs best on Asphalt3k and Concrete3k. On Asphalt3k, ECFormer-B48 achieves the highest mIoU of 81.29 % and an F1 score of 88.65 %, outperforming all comparison models. On Concrete3k, ECFormer-B48 leads with an mIoU of 86.36 % and an F1 score of 93.83 %, again the highest among the models compared.

Table 4
Performance comparisons on Asphalt3k and Concrete3k.

Model	Asphalt3k		Concrete3k	
	mIoU (%)	F1 (%)	mIoU (%)	F1 (%)
UNet	78.52	86.93	84.20	91.05
OCRNet	79.84	87.05	85.61	92.11
DMA-Net	80.23	87.22	84.46	91.37
CrackFormer	80.43	87.63	85.67	92.82
SegCrack	80.32	88.09	85.59	92.66
ECFormer-B48	81.29	88.65	86.36	93.83

In contrast, OCRNet's performance on Concrete3k, with an mIoU of 85.61 % and an F1 score of 92.11 %, while commendable, still falls short of ECFormer-B48's performance. Other advanced models specifically designed for crack segmentation, such as DMA-Net, CrackFormer, and SegCrack, also perform inferiorly to ECFormer-B48 across both datasets.

Fig. 11 displays some visual results of compared models on Asphalt3k, including ECFormer-B48, UNet, OCRNet, DMA-Net, CrackFormer, and SegCrack. In the first row, which shows global asphalt cracks, ECFormer-B48 accurately captures the overall structure and details of the cracks, outperforming UNet and DMA-Net, which exhibit significant omissions. In the second row, depicting low-contrast discontinuous cracks, ECFormer-B48, despite some false detections, still surpasses other models. In the third row's complex scenes with tiny cracks, ECFormer-B48 precisely identifies cracks that are difficult to capture, whereas UNet, CrackFormer, and SegCrack severely miss such features. The fourth row, featuring cracks on white lane markings, shows ECFormer-B48's robustness despite some omissions, outperforming other models that misidentify features.

Fig. 12 illustrates some visual results on Concrete3k. In the first row, images fraught with severe impact noise typically induce false positives in compared models; however, ECFormer-B48 demonstrates superior performance by accurately delineating cracks without succumbing to these errors. The cracks in the second row have different scales, one thick and one thin. Here, ECFormer-B48 stands out by successfully detecting the finer cracks, an ability that eludes other models, thus showcasing its acute sensitivity to minute structural details. In the third row, the focus shifts to intermittent cracks, which pose significant difficulties. Both ECFormer-B48 and OCRNet, leveraging their high-resolution strengths, adeptly identify these elusive features, unlike DMA-Net, which largely fails in this regard. The fourth row tests the models' ability to distinguish between genuine cracks and similar-looking linear patterns. While OCRNet, DMA-Net, and SegCrack misclassify these patterns as cracks, our proposed method effectively resists such misidentifications, highlighting its exceptional specificity and robustness against false positives.

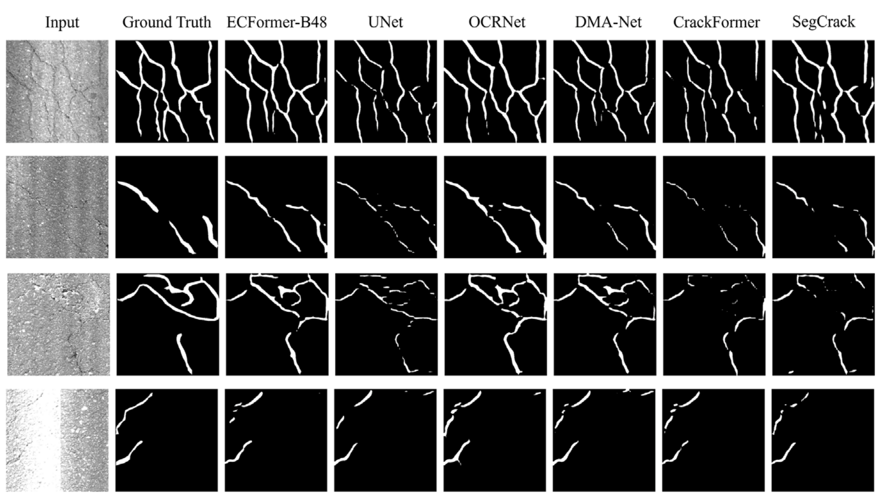


Fig. 11. Qualitative results of ECFormer-B48, UNet, OCRNet, DMA-Net, CrackFormer and SegCrack on Asphalt3k.

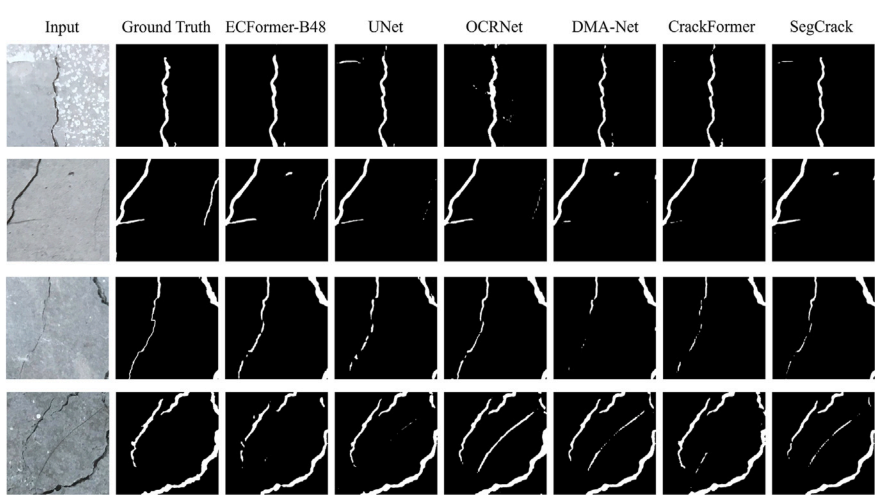


Fig. 12. Qualitative results of ECFormer-B48, UNet, OCRNet, DMA-Net, CrackFormer and SegCrack on Concrete3k.

Table 5
Robustness on CrackSeg9k (400 × 400). Moderate severity (s=3) mIoU per corruption and aggregate robustness score RS.

Model	Clean	Noise	Motion	Brightness	JPEG	RS
OCRNet (HRNet-W18)	80.9	75.2	72.8	77.1	76.0	91.8
DMA-Net (ResNet18)	80.7	74.7	71.2	76.7	75.7	90.9
CrackFormer	80.2	74.5	71.2	76.8	74.8	91.2
SegCrack	79.6	73.6	71.5	75.6	74.1	91.2
ECFormer-B48	**81.8**	**76.8**	**73.8**	**78.5**	**76.6**	**92.2**

Overall, ECFormer exhibits exceptional performance across various crack scenarios, precisely capturing the boundaries and shapes of tiny cracks, demonstrating its potential and superiority in crack segmentation tasks.

5.4. Robustness to common corruptions

Following the corruption taxonomy and severity design popularized by ImageNet-C [89], four common corruptions are adapted to semantic segmentation in the crack domain: Gaussian noise, motion blur, brightness (illumination) shift, and JPEG compression. Each corruption is evaluated at five severities (s=1–5). Unless stated, tables report the moderate level (s=3), and figures present full severity curves. All evaluations are conducted on CrackSeg9k with batch= 1, input 400×400, and no test-time augmentation. Formal operator definitions are provided in Appendix A.

Let m_{clean} denote the mIoU on clean images and $m_{p,s}$ the mIoU under corruption family p at severity s. The relative drop is defined as

$$\Delta_{p,s} = \frac{m_{\text{clean}} - m_{p,s}}{m_{\text{clean}}} \times 100\ \%. \tag{21}$$

Robustness is summarized by the aggregate score

$$\text{RS} = 100 - \frac{1}{\sum_p |S_p|} \sum_p \sum_{s \in S_p} \Delta_{p,s}, \tag{22}$$

where higher RS indicates a smaller average relative drop across families and severities.

Severity levels for the four corruptions are listed in Table A.1. All operators act on per-channel normalized images in $[0, 1]$; formal definitions appear in Eqs. (A.1) to (A.4).

ECFormer-B48 achieves the best overall robustness (RS = 92.2), ahead of OCRNet (91.8) and DMA-Net (90.9) in Table 5. At the moderate level (s=3), ECFormer-B48 consistently tops all four families (Noise 76.8, Motion 73.8, Brightness 78.5, JPEG 76.6), with margins typically + 0.6–2.0 mIoU over strong CNN and Transformer baselines. The sensitivity ordering is stable across models—JPEG and motion blur incur the largest losses, whereas brightness is comparatively benign. Curves over s=0–5 in Fig. 13 confirm near-monotonic degradation with ECFormer-B48 retaining the leading trajectory up to s=5.

5.5. Ablation study

To rigorously evaluate the individual contribution of each proposed module (MSSRA, ERLA, and BDCRFE) to the overall performance and efficiency gains, we conduct a series of detailed ablation studies in this section. The ablation experiments are performed on CrackSeg9k, where the model is trained from scratch using Kaiming Normal initialization [90]. To evaluate the performance of the proposed modules, convolution block with similar parameters is used to replace each proposed attention module, with the results of this model serving as the baseline for comparative experiments. Subsequent comparative models using convolution are referred to as baseline models. For efficient ablation analysis, only ECFormer-B16 is used, with image cropping set to 200×200 and iterations limited to 20,000.

Initially, the convolutional blocks in the baseline are sequentially replaced with MSSRA, ERLA, and BDCRFE. It is important to note that MSSRA replaces the convolution block only in the HR branch, while ERLA is exclusively used in the LR branch. The results are shown in Table 6. By substituting the convolution with MSSRA, the model's mIoU increased to 77.60 %, showing an improvement of 0.56 %. Although MSSRA introduces additional computational costs, increasing

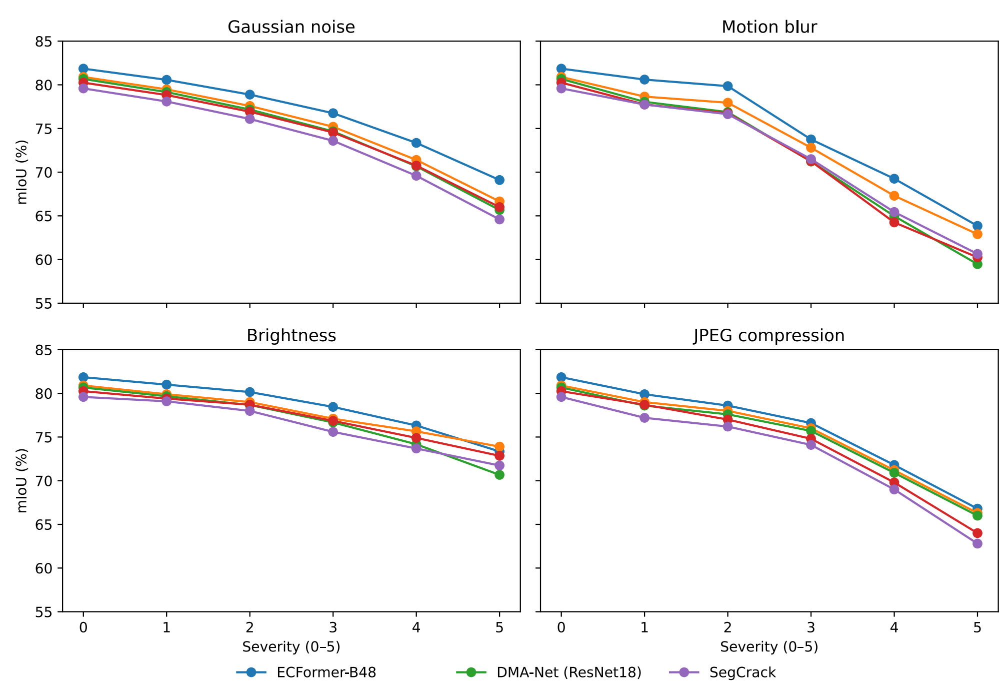


Fig. 13. Robustness on CrackSeg9k: mIoU vs. severity (s=0–5) across four corruption families (Gaussian noise, Motion blur, Brightness, JPEG) for five models (OCRNet, DMA-Net, CrackFormer, SegCrack, ECFormer-B48).

Table 6
Ablation experiments on CrackSeg9k.

Model	mIoU (%)	Params (M)	GFLOPs	FPS
Baseline	77.04	0.78	0.144	212.2
+ MSSRA	77.60	0.79	0.147	185.1
+ ERLA	77.83	0.80	0.162	163.8
+ BDCRFE	78.05	0.82	0.185	143.7

Table 7
Performance comparison of different attention mechanisms in the HR branch.

Module	mIoU (%)	Params (M)	GFLOPs	FPS
Conv	77.04	0.78	0.144	212.2
MSA	77.65	0.79	0.742	62.5
SRA ($p = 7$)	77.48	0.79	0.154	160.7
ERLA	77.45	0.79	0.144	192.0
MSSRA ($p = 1, 3, 5, 7$)	77.60	0.79	0.147	185.1

the GFLOPs to 0.147 and slightly decreasing the FPS to 185.1, it improves the performance by adding very few parameters. After replacing the convolution with ERLA, the mIoU reached 77.83 %, a rise of 0.79 % over the baseline. ERLA increased the computational costs to 0.162 GFLOPs and decreased the FPS to 163.8. Replacing the convolution with BDCRFE further improved the mIoU to 78.05 %. The introduction of BDCRFE slightly increased the parameters to 0.82 M and the computational costs to 0.185 GFLOPs, with a frame rate of 143.7 FPS. BDCRFE's bidirectional attention enhanced feature representation across different branches, significantly boosting the model's performance. It is worth noting that the introduction of the proposed efficient attention module only slightly increases the model's parameters and adds only a controllable computational overhead. This is because the attention mechanism involves only two learnable linear projections.

5.5.1. Impact of MSSRA

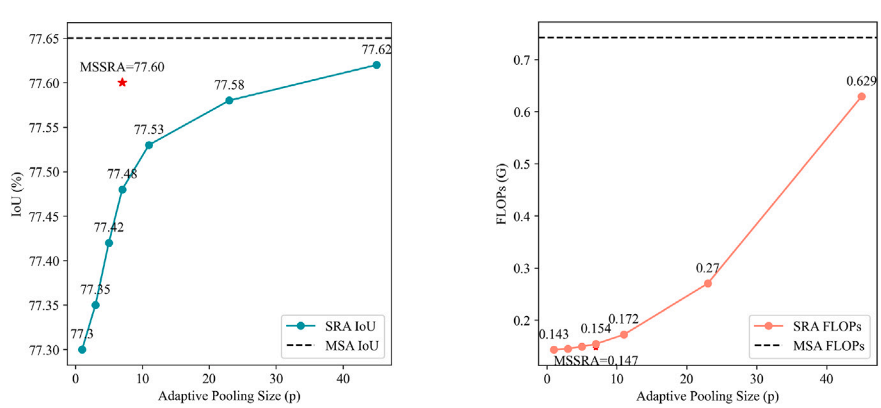
To validate the effectiveness of MSSRA, the original MSA [9], SRA [73], and the proposed MSSRA are each used to replace the convolution in the HR branch of the baseline model. The results are shown in Table 7, where Conv represents the baseline. In SRA and MSSRA, p represents the adaptive pooling size in the spatial reduction module. In all configurations, the number of heads in multi-head attention is set to 4.

From Table 7, it can be observed that, compared to the baseline model using only convolution, the introduction of MSA improved the model's mIoU, increasing from 77.04 % to 77.65 %. However, this improvement comes at the cost of significantly higher computational complexity. Specifically, the baseline model requires only 0.144 GFLOPs, while replacing it with MSA increases the computation by five times, reaching 0.742 GFLOPs. Given the modest performance improvement, this trade-off is inefficient.

The improvement made by SRA to MSA is immediate, as it significantly reduces computational cost from 0.742 to 0.154 GFLOPs with the introduction of very few additional parameters. However, compared to MSA, SRA does not significantly improve the performance over the baseline model, indicating that SRA cannot fully exploit the potential advantages of multi-head attention. This paper argues that the main difference between SRA and MSA lies in SRA's reduction of the spatial resolution of the Key and Value, which is set to a fixed scale as 7×7 in PVTv2 [73]. This setup reduces the model's ability to capture multi-scale information. Therefore, this paper compares SRA with different scales to observe the impact of spatial reduction on the self-attention mechanism. Specifically, the model's performance is compared with reduction factors p set to 1, 3, 5, 7, 11, 23, and 45, as shown in Fig. 14.

In Fig. 14(a), it can be observed that the performance of the SRA gradually approaches that of the MSA as p continues to increase, but the magnitude of the increase gradually tapers off. In addition, comparing in Fig. 14(b), it is further observed that the increase in p leads to a much larger increase in computational overhead than the performance gain it brings. Based on the above observations, this paper follows the setup of PVTv2 [73], but limits the maximum adaptive pooling size to $7{\times}7$, while setting three smaller scales of 1, 3, and 5. Each spatial reduction operation is applied to a different attention head, with the results shown in the last row of Table 7. To further validate our choice of the multi-scale pooling configuration for MSSRA, we also conducted a sensitivity study. Based on our findings in Fig. 14(b), which indicate that pooling sizes larger than 7 offer diminishing performance returns at a significantly higher computational cost, our analysis of different configurations focused on a range of more efficient pooling sizes. The results in Table 8 demonstrate the superiority of our balanced {1,3,5,7} configuration. We found that alternative setups, which either lacked sufficient global context (e.g., {4,5,6,7}) or failed to preserve fine-grained spatial details (e.g., {1,2,3,4}), both resulted in a drop in performance. This confirms that an effective multi-scale approach must capture a wide spectrum of features, from global context to local details, and our proposed configuration strikes the optimal trade-off.

Compared to SRA, MSSRA further reduces computational costs while maintaining the same level of inference efficiency, but its performance is closer to MSA. Taking the convolution-based baseline model and an MSA-based model as the upper and lower bounds for comparison, it can be concluded that SRA significantly improves computational efficiency by reducing computational costs, while the proposed MSSRA, by



(a) (b)

Fig. 14. Effect of pooling size p on IoU and FLOPs.

Table 8
Sensitivity analysis of different pooling configurations for MSSRA.

Configuration	mIoU(%)	GFLOPs
p =1, 2, 3, 4	77.25	0.145
p =4, 5, 6, 7	77.18	0.152
p =1, 5, 9, 13	77.51	0.192
p =1, 3, 5, 7	**77.60**	**0.147**

Table 9
Performance comparison of different attention mechanisms in the LR branch.

Module	mIoU (%)	Params (M)	GFLOPs	FPS
Conv	77.04	0.78	0.144	212.2
MSA	77.60	0.79	0.516	105.3
RLA	77.47	0.79	0.148	195.8
MSSRA($p = 1, 3, 5, 7$)	77.36	0.79	0.145	201.6
ERLA	77.58	0.79	0.158	189.1

introducing multi-scale learning capability, achieves a better balance between performance and efficiency compared to SRA.

Additionally, the performance of ERLA in the HR branch is compared. ERLA's performance is close to SRA but inferior to the proposed MSSRA, and such results are related to the different roles of the branches. The HR branch maintains a fixed resolution throughout, mainly focusing on capturing intricate edges and texture information in crack images. The proposed MSSRA, through multi-scale pooling operations, enables the model to simultaneously focus on features at different scales. By applying spatial reduction operations at various scales across multiple self-attention heads, it preserves key information at different scales. Consequently, the model can capture both large-scale global structures and small-scale crack details. In contrast, ERLA is primarily designed to enhance global modeling capability in a large receptive field, which explains its weaker performance in the HR path compared to MSSRA. This validates the rationale for using MSSRA in the HR branch.

5.5.2. Impact of ERLA

The original MSA [9], the original RLA [68], and the proposed ERLA are replaced with the convolution of the LR branch in ECFormer to explore their effects on the model performance, and the results are shown in Table 9.

From the experiment, a phenomenon similar to the HR branch can be observed. The introduction of MSA effectively improves performance, but the computational cost is 3.6 times higher than the baseline model, increasing from 0.144 to 0.516 GFLOPs. In contrast, RLA improves performance over the baseline model, while the computational overhead is drastically reduced from 0.516 GFLOPs to 0.148 GFLOPs compared to the MSA, suggesting that RLA can control the consumption of computational resources more effectively. However, the performance improvement of RLA relative to the baseline is not as substantial as that of MSA. The introduction of ERLA achieves a better balance between performance and computational cost. Specifically, ERLA improves mIoU from 77.47 % to 77.58 %, surpassing RLA but slightly lower than MSA. In terms of computational cost, ERLA remains at the same level as the baseline model and is far lower than MSA, reducing from 0.516 to 0.158 GFLOPs. Such results show that ERLA is a more practical and efficient choice, as it maintains high performance while avoiding the excessive computational resource consumption associated with MSA.

The result of applying MSSRA to the LR branch is also compared. The main role of MSSRA is to enhance the model's ability to learn at multi scales, whereas the LR branch is designed to learn global features as a complement to the HR branch. As observed in Table 9, ERLA achieves better performance than MSSRA, which confirms the earlier analysis that a specific branch serves distinct capability.

Table 10
Performance comparison of different fusion methods.

Module	mIoU(%)	Params(M)	GFLOPs	FPS
Conv	77.04	0.78	0.144	212.2
H2LCA (MSA)	77.53	0.79	0.545	92.7
H2LCA (MSSRA)	77.49	0.79	0.147	183.8
L2HCA (MSA)	77.28	0.79	0.544	95.5
L2HCA (ERLA)	77.20	0.79	0.158	187.2
BDCRFE (MSA)	77.68	0.81	0.982	43.6
BDCRFE (MSSRA, ERLA)	77.62	0.81	0.165	170.1

5.5.3. Impact of BDCRFE

Global and detailed features are two key attributes for representing crack images and are crucial for achieving high-performance semantic segmentation. Relying solely on one type of feature for prediction leads to suboptimal results. The proposed BDCRFE treats the HR and LR branches as features from different modalities and establishes an effective communication channel between the branches using cross-attention.

Table 10 presents the results of different feature fusion methods, where Conv indicates the use of element-wise addition in the baseline model [50], H2LCA indicates that Query in cross-attention is from the HR branch and Key and Value are from the LR branch, and L2HCA is the opposite. It can be observed that all attention-based fusion methods outperform the baseline, demonstrating that element-wise addition fusion is insufficient. When comparing the fusion methods of H2LCA and L2HCA, H2LCA consistently outperforms L2HCA, indicating that the HR branch has a greater impact on the model's performance. BDCRFE surpasses both H2LCA and L2HCA, suggesting that the bidirectional use of cross-attention more effectively leverages the feature information from different branches during HR and LR feature fusion. Specifically, BDCRFE with MSA improves mIoU performance by 0.15 % and 0.4 % compared to H2LCA and L2HCA, respectively.

In addition to the original MSA, the proposed MSSRA and ERLA are also compared here. Specifically, MSSRA and ERLA are substituted for the MSA in H2LCA and L2HCA. As observed in Table 10, the results are consistent with the previous Sections 5.5.1 and 5.5.2. The proposed efficient attention modules effectively reduce computational costs, but the model's mIoU only shows a slight decrease, from 77.68 % to 77.62 %. Considering the significant improvement in model inference speed, from 43.6 FPS to 170.1 FPS, such trade-off offers a highly cost-effective balance.

6. Limitations and future work

Although the proposed ECFormer performs well across multiple datasets, it is important to acknowledge its limitations and outline directions for future research.

First, concerning computational efficiency and deployment scenarios, while ECFormer demonstrates a superior trade-off between performance and efficiency compared to other state-of-the-art models, its Transformer-based architecture is inherently more complex than specialized lightweight CNNs. This may pose challenges for deployment in extremely resource-constrained environments, such as on embedded or edge devices. Therefore, a direct comparative study against architectures specifically designed for mobile platforms (e.g., MobileNet-based segmentation models) would be a valuable future investigation to fully characterize the performance-versus-resource trade-offs. Furthermore, exploring model compression techniques, such as quantization and pruning, is another promising direction to adapt ECFormer for edge computing applications.

Second, regarding model robustness and specific failure modes, we observed that ECFormer's performance can deteriorate when facing significant image quality degradation, such as severe noise and lighting changes found in the Asphalt3k and Concrete3k datasets. More specifically, our qualitative analysis indicates that the model may occasionally fail to detect extremely fine or discontinuous cracks, or misclassify complex background textures that mimic crack patterns. While ECFormer generally shows strong resilience, future work should include more systematic robustness testing against controlled noise, lighting variations, and other corruptions. Developing mechanisms to specifically enhance the detection of these challenging, fine-grained crack features is another critical research avenue.

Finally, in terms of generalization and data dependency, the model's performance was validated on three comprehensive datasets, but its generalizability to unseen crack morphologies or different material types remains to be further explored. This limitation is tied to a broader challenge in the field: the scarcity of large-scale, high-quality annotated crack datasets. To this end, a promising future direction is to leverage parameter-efficient fine-tuning methods, such as Adapters [91], to transfer knowledge from large-scale pre-trained vision models (e.g., SAM [92] and CLIP [93]) to the crack segmentation task. This could significantly enhance model generalization and robustness without requiring massive task-specific datasets.

7. Conclusions

Research on Transformer-based crack recognition is gaining increasing attention, as its self-attention mechanism, when applied to images, naturally possesses long-range modeling capability and a global receptive field. However, its quadratic complexity with respect to the input sequence results in computational costs that are difficult to accept when processing high-resolution images. To address this issue, our paper introduces three efficient attention modules and designs an efficient crack segmentation model, ECFormer. For the HR branch, which primarily learns the local features of cracks, a multi-scale spatial-reduction attention module is designed. This module significantly reduces computational complexity by performing spatial reduction on feature maps at multiple scales while adapting to the feature extraction needs of targets at different scales, allowing the model to excel in capturing multi-scale features. For the LR branch, which primarily learns global semantic features, an enhanced ReLU-based linear attention module is designed. While maintaining a global receptive field, this module reduces the computational complexity to a linear level. The enhanced linear attention module improves feature discrimination and model expressiveness by adjusting the direction of each Query and Key, bringing similar features closer together and pushing dissimilar features further apart. To achieve efficient fusion of HR and LR features, a bidirectional cross-resolution feature enhancer is designed. This module establishes an effective feature exchange channel between the HR and LR branches through a bidirectional cross-attention mechanism, enhancing the model's feature fusion and overall performance. Comprehensive evaluations on general crack datasets and specific scenario crack datasets show that ECFormer consistently outperforms other comparison models. To promote advancements in the field of crack recognition, the data and code from this paper will be made publicly available at https://github.com/CHDyshli/efficient_crack_transformer.

CRediT authorship contribution statement

Yongshang Li: Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Zihan Ma:** Visualization, Validation. **Ronggui Ma:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table A.1
Severity parameterization for four corruptions on CrackSeg9k. All operators act on per-channel normalized images in $[0,1]$; formal definitions are given in Eqs. (A.1)–(A.4).

Corruption	s=1	s=2	s=3	s=4	s=5
Gaussian noise σ (× 1/255)	5	10	15	20	25
Motion blur length L (pixels; random angle)	3	5	7	9	11
Brightness shift (additive, %)	±10	±20	±30	±40	±50
JPEG quality Q	90	70	50	30	10

Appendix A. Robustness corruptions for CrackSeg9k: definitions and parameters

This appendix formalizes the four corruption operators used in Section 5.4: Gaussian noise, motion blur, brightness shift, and JPEG compression. Parameter values follow the five severity levels (s=1–5) in Table A.1. The operators correspond one-to-one to the columns used in the results of Section 5.4.

Let $x \in [0,1]^{H\times W\times C}$ be the input image. Unless stated, evaluation uses batch= 1, input 400×400, no test-time augmentation. Randomness arises from Gaussian noise and the motion-blur angle; results are averaged over N repeats with fixed seeds. All operators act in the $[0,1]$ domain and end with clipping to $[0,1]$. Convolution uses bilinear sampling for subpixel offsets and `reflect` padding at borders.

Gaussian noise (additive). The operator adds i.i.d. Gaussian noise to each channel:

$$x' = \mathrm{clip}\big(x + (\sigma/255)\epsilon, 0, 1\big), \qquad \epsilon \sim \mathcal{N}(0, I). \tag{A.1}$$

Applied independently per channel. Severity mapping: $\sigma \in \{5, 10, 15, 20, 25\}$ corresponds to $\{5, 10, 15, 20, 25\}/255$.

Motion blur. The operator convolves the image with a normalized line kernel:

$$x' = x * k_{L,\theta}, \tag{A.2}$$

where $*$ denotes channel-wise 2D convolution and $k_{L,\theta}$ is a line point-spread function of length L at angle θ. The angle θ is drawn uniformly from $[0,\pi)$ once per image. Convolution uses bilinear sampling for sub-pixel offsets and `reflect` padding at borders. Severity mapping: $L \in \{3, 5, 7, 9, 11\}$ (pixels).

Brightness shift. The operator adds a constant offset to all channels:

$$x' = \mathrm{clip}\big(x + \beta, 0, 1\big). \tag{A.3}$$

The per-image constant β is applied uniformly to all channels. Severity mapping: $\beta \in \{\pm 0.10, \pm 0.20, \pm 0.30, \pm 0.40, \pm 0.50\}$.

JPEG compression. The operator applies a codec round trip in the 8-bit domain:

$$x' = \mathrm{JPEG}^{-1}\big(\mathrm{JPEG}(\mathrm{round}(255x), Q)\big)/255, \tag{A.4}$$

using a standard JPEG implementation with YCbCr 4:2:0 subsampling. Severity mapping: $Q \in \{90, 70, 50, 30, 10\}$ (quality factor; lower is stronger).

Data availability

Data will be made available on request.

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